

Novelty & Differentiation Report

E-Commerce Product Quality & Customer Pain-Point Intelligence

Core Strategy Statement: We do not claim absolute unprecedented invention ("Nobody has done this before"). Instead, our scientific contribution lies in constructing a current, large-scale web-scraped dataset and integrating predictive analytics with multi-indicator scoring frameworks to expose hidden product deficiencies.

1. Introduction & Contextual Positioning

E-commerce platforms provide a wealth of structured product information—such as pricing, discounting structures, aggregate ratings, review counts, and categorical identifiers—alongside unstructured customer feedback (reviews and text queries). While analyzing sentiment or predicting numerical ratings are established baseline areas, treating them in isolation misses critical operational flaws. This report outlines our structured methodology for combining structured metadata with advanced NLP and textual mining to build an actionable product intelligence engine.

2. Detailed Novelty Features

Novel Feature 1: Product Health Score (PHS)

Traditional metrics rely solely on star ratings, which are frequently biased or gamed. The Product Health Score (PHS) synthesizes numerical ratings, NLP-derived review sentiment velocity, review volume confidence weights, price positioning, discount depth, and negative-topic frequency into a composite index. Products are automatically classified into three actionable operational tiers:

- **Strong:** High verified sentiment, low negative frequency, robust review volume, and healthy price retention.
- **Watchlist:** Moderate indicators, minor emerging complaints regarding packaging or minor features requiring monitoring.
- **At-Risk:** High ratings masking high negative-topic frequencies or recurring structural complaints.

Novel Feature 2: Rating-Review Discrepancy / Hidden Pain-Point Score

A major vulnerability in modern e-commerce is the *"Paper Tiger"* phenomenon—products that maintain a high numerical rating (e.g., 4.5/5) but suffer from severe recurring negative themes buried deep within textual customer reviews. Our framework computes a Discrepancy Index by contrasting quantitative star scores against semantic polarity distributions, isolating items where customer sentiment directly contradicts nominal ratings.

Novel Feature 3: Category-Specific Pain Points

Customer dissatisfaction drivers vary wildly across product domains (e.g., electronics failures center on battery degradation and software bugs, whereas apparel complaints focus on sizing accuracy and fabric durability). Rather than applying a blanket text-mining model, our pipeline dynamically compares what drives negative customer experience across diverse categories using custom-extracted corpus taxonomies.

Novel Feature 4: Price-Sentiment-Rating Intelligence

By mapping price points and discount percentages against review volume, sentiment polarity, and ratings, this feature categorizes merchandise into positioning segments. It identifies whether aggressive discounting correlates with diminished build quality perception or whether premium pricing successfully insulates products from customer churn.

3. E-Commerce Differentiation Matrix

The matrix below highlights how our framework shifts focus from common, elementary analytical tasks to robust diagnostic business intelligence:

Dimension	Existing / Common Focus	Our Contribution
Sentiment	Binary positive/negative classification of reviews.	Use granular sentiment decomposition to actively diagnose hidden product weaknesses.
Rating	Static numerical rating prediction models.	Compare numerical star ratings directly against textual pain-point frequencies to uncover "Paper Tigers".
Products	Collaborative filtering recommendation or association mining.	Dynamic Product Health and At-Risk operational classification.
Categories	Aggregate platform-wide performance tracking.	Category-specific customer pain-point intelligence and root-cause taxonomies.
Business Output	Basic model accuracy metrics and visual charts.	Actionable product improvement, quality control, and pricing positioning recommendations.

4. Systematic Novelty Validation Matrix

Comparing our project framework against representative baseline studies demonstrates our unique integration of structured data, text mining, and business metrics:

Study / Framework	Dataset	Prediction	Text Mining	Association	Price/Discount	Business Metric	Our Difference
Study 1 (Baseline)	Static CSV Archive	Rating (RMSE)	Basic TF-IDF	None	Ignored	Accuracy Score	Static review focus only
Study 2 (Baseline)	Single Category	Polarity Class	Lexicon VADER	Market Basket	Included	Sales Volume	No pain-point diagnosis
Study 3 (Baseline)	Multi-category API	Sales Forecast	Topic Modeling	Frequent Items	Included	Revenue Projections	Lacks hidden defect scoring
Our Project	Fresh Scraped Data	Health & Risk	Advanced NLP	Included	Integrated	Product Health Score	Integrated Analytical Framework

5. Research Questions & Business Output

Core Research Question: *"Which products appear successful from surface-level numerical ratings, but contain hidden recurring customer problems, and what are their underlying root causes?"*

Operational Output: The analytical output translates directly into manufacturer quality-control interventions, inventory risk mitigation for platform operators, and precise consumer buying guides.

6. Data Collection & Preprocessing Plan

- Pipeline & Source:** Automated web-scraping pipeline targeting major retail catalogs across 5 distinct consumer goods categories.
- Collection Parameters:** Extraction timeframe spanning [Collection Dates Placeholder], capturing product IDs, titles, list prices, promotional discounts, aggregate star ratings, review timestamps, and full textual review bodies (Target: \$N \ge 15,000\$ unique items with multi-review histories).
- Data Cleaning & Preprocessing:** Handling missing values, filtering bot/spam reviews using linguistic entropy measures, tokenization, stop-word removal, and lemmatization for aspect-based sentiment analysis.